

Note 8: Note of semisimple ring and some exercises

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8.1 Summary of Progress

1. Delivered a three hours lecture in algebra taking group about representation theory: semisimple ring.
2. Wrote some notes on this topic and recorded some video of the speech.

8.2 Note of group representation theory: semisimple ring

Let R be a ring (not necessary commutative), G is a finite group in the section. The proof of the results below refer to book[2] in chapter C.2.

8.2.1 the properties of semisimple ring

Definition 1 (minimal left ideal) The Jacobson radical $J(R)$ is defined to be the intersection of all the maximal left ideals of R .

Definition 2 (minimal left ideal) We say left ideal L of R is minimal if there is no left ideal I of R such that $0 \subsetneq I \subsetneq L$.

Proposition 3 We have

- $J(R) = \bigcap_{I=\text{maximal left ideal}} \text{ann}(R/I)$, so it is an two-sided ideal of R .
- Moreover,

$$\begin{aligned} J(R) &= \{x \in R : \forall r \in R, \exists u \in R, u(1 - rx) = 1\}. \\ &= \{x \in R : \forall r, s \in R, 1 - rxs \in U(R)\}. \end{aligned}$$

Example 4 .

1. If $R = \mathbb{Z}$, we have $J(R) = 0$.
2. If $R = \text{Mat}_n(k)$ for a division ring k , then

$$\text{COL}(l) := RE_{11} = \{[a_{ij}] \in R : a_{ij} = 0 \text{ if } j \neq l\}$$

is a minimal left ideal of R . Moreover,

$$\text{COL}^*(l) := \bigoplus_{j \neq l} \text{COL}(j)$$

is a maximal ideal of R , thus $J(R) = 0$.

Definition 5 For R -module $M \neq 0$, we say M is

- **simple** if M has only two submodules $0, M$.
- **semisimple** if M is a direct sum of simple modules.

Theorem 6 An R -module M is semisimple if and only if for any submodule N of M is a direct summand, i.e. there is a submodule N' such that $M = N \oplus N'$.

Definition 7 For $R \neq 0$, we say R is

- **simple** if R has only two two-sided ideal I , i.e. $I = 0, R$.
- **left semisimple** if R is a direct sum of minimal left ideals.
- **right semisimple** if R is a direct sum of minimal right ideals.

Remark 8 .

A simple module is necessary a semisimple module, but a simple ring is **not** necessary a left semisimple ring. For example, let $R = \text{End}_{\mathbb{R}} V$ with V is an infinite dimensional vector space over $\mathbb{R}$, then R is a simple ring but not a left or right semisimple ring.

Remark 9 . If R is left semisimple, then R is necessary a direct sum of **finitely many** minimal left ideals. i.e. $R = L_1 \oplus \dots \oplus L_n$.

Theorem 10 (Nakayama's theorem) If A is a finitely generated left R -module and $J(R)A = A$, then $A = 0$.

Definition 11 For $R \neq 0$, we say R is

- **left noetherian** if R has ACC on left ideal: every ascending chain of left ideals $I_1 \subseteq I_2 \subseteq I_3 \subseteq \dots$ stops;
- **left artinian** if R has DCC on left ideal: every descending chain of left ideals $I_1 \supseteq I_2 \supseteq I_3 \supseteq \dots$ stops.

Theorem 12 R is left semisimple $\iff J(R) = 0$ and R is left artinian.

Theorem 13 The following statements are equivalent:

1. R is left semisimple.
2. Every left R -module is semisimple.
3. Every left R -module is injective.
4. Every left R -module is projective.
5. Every short exact sequence of left R -module splits.

Let G is a group and k is a field. Let $(a_g)_{g \in G} \in k^{\oplus |G|}$, and denote $(a_g)_{g \in G} = \sum_{g \in G} a_g g$. And define

$$\left(\sum_{g \in G} a_g g \right) \left(\sum_{h \in G} b_h h \right) = \sum_{g, h \in G} a_g b_h gh,$$

then kG is a k -algebra, called the group algebra of G over k .

Theorem 14 (Maschke) If G is a finite group, and k is a field with $\text{char } k$ does not divide $|G|$, then kG is left semisimple.

Proof . For any kG -modules M , it is suffices to prove M is semisimple.

- Let N is a submodule, then N is a vector space of k , thus there is projector $P : M \rightarrow N$. Define

$$\bar{P} : M \rightarrow N, \quad \bar{P}(u) = \frac{1}{|G|} \sum_{g \in G} g^{-1} P(gu).$$

- Verifies that $\bar{P}$ is an kG -map:

$$\bar{P}(hu) = \frac{1}{|G|} \sum_{g \in G} g^{-1} P(ghu) = \frac{1}{|G|} \sum_{y=gh \in G} (yh^{-1})^{-1} P(yu) = \frac{1}{|G|} \sum_{y \in G} hy^{-1} P(yu) = h\bar{P}(u).$$

thus $\text{Ker } \bar{P}$ is a submodule of M .

- Since $\bar{P}^2 = \bar{P}$, $M = \text{Im } \bar{P} \oplus \text{Ker } \bar{P}$.
- By theorem 6, M is a semisimple kG -module.

By theorem 13, kG is left semisimple. ■

8.2.2 The structure of left semisimple ring

Theorem 15 (Shur lemma) .

- If M, M' are simple modules, then $f : M \rightarrow M'$ is a zero map or an isomorphism.
- If M is a simple modules, then $\text{End}_R M$ is a division ring. Moreover, $\text{End}_R L$ is a division ring if L is a minimal left ideal.

Lemma 16 If R is left semisimple, and M is a simple R -module, then $M \simeq L$, a minimal left ideal of R . Thus

$$\{\text{simple } R\text{-modules}\} / \cong = \{\text{minimal left ideals of } R\} / \cong$$

Proof Since $R = L_1 \oplus \dots \oplus L_n$, we have

$$M \cong \text{Hom}_R(R, M) = \text{Hom}_R\left(\bigoplus L_i, M\right) = \bigoplus \text{Hom}_R(L_i, M) \neq 0$$

Thus $\text{Hom}_R(L_i, M) \neq 0$ for some i , then $L_i \simeq M$. ■

Theorem 17 (Wedderburn-Artin) .

1. R is left semisimple $\iff R$ is a finite direct sum of matrix rings over division rings.
2. $R = L_1 \oplus \dots \oplus L_n$ is a decomposition of left ideals, then

$$R \simeq \text{Mat}_{n_1}(k_1) \times \dots \times \text{Mat}_{n_r}(k_r), \quad k^{\text{op}} \simeq \text{End}_R L_i.$$

Moreover, the numbers n_i, r as well as the division rings k are uniquely determined by R .

Remark 18 R is left semisimple $\iff R$ is right semisimple

Corollary 19 If

- k is an **algebraic closed field** ;
- G is a **finite group** ;

- $\text{char } k$ does not divide $|G|$;

Then

$$kG \simeq \text{Mat}_{n_1}(k) \times \dots \times \text{Mat}_{n_r}(k).$$

Remark 20 If kG is left semisimple, then

$$\{\text{simple } kG\text{-modules}\} / \cong \xleftrightarrow{1:1} \{\text{minimal left ideals of } kG\} / \cong$$

Corollary 21 If

- k is an **algebraic closed field** ;
- G is a **finite group** ;
- $\text{char } k$ does not divide $|G|$;

Then

$$|G| = n_1^2 + \dots + n_r^2,$$

where

- $r = \# \{\text{minimal left ideals of } kG\} / \cong$
- $n_i = \dim L_i$, L_i is a representative elements of $\{\text{minimal left ideals of } kG\} / \cong$.

8.3 Plan for next week

1. Textbook[1] chap.5.
2. Textbook [2] C2.7-2.9, the character of representation.

References

- [1] M. A. Nielsen and I. L. Chuang. *Quantum Computation and Quantum Information*, 10th Anniversary Edition. Cambridge University Press, 2010.
- [2] J. J. Rotman. *Advanced Modern Algebra*, Third edition. Graduate Studies in Mathematics, Volume 165. American Mathematical Society, 2015. ISBN 978-1-4704-1554-9 (pt. 1), 978-1-4704-2311-7 (pt. 2).